Measure Theoretic Probability
Lesson 17
General Random Variables and Borel Sets on $\mathbb{R}^k$

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Outline

1. Measurable Maps and Random Variables
2. The Generating-Class Lemma
3. A Practical Test for Random Variables
4. $\sigma(X)$ for General Random Variables
5. Random Vectors and Borel Sets on $\mathbb{R}^k$
6. When Is a Vector a Random Vector?
7. Summary and Exercises

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Measurable Maps Between Measurable Spaces

Setting

Let Ω_1, Ω_2 be nonempty sets with σ-fields $\mathcal{F}_1, \mathcal{F}_2$. These are **measurable spaces**: no measures are needed (though they could carry μ_1, μ_2).

Definition

A function $f : \Omega_1 \to \Omega_2$ is **measurable** $\mathcal{F}_1/\mathcal{F}_2$ if

$$f^{-1}(B) \in \mathcal{F}_1 \qquad \text{for every } B \in \mathcal{F}_2.$$

Remark

We have used this idea before and simply said "f is measurable". Because measurability is tied to the σ-fields, we write "measurable $\mathcal{F}_1/\mathcal{F}_2$" whenever there could be confusion.

Random Variables and the Borel σ-Field

Random variable

Let $(\Omega, \mathcal{F})$ be a measurable space (add a probability measure $\mathbb{P}$ if you like; it is not needed here). A function $X : \Omega \to \mathbb{R}$ is a **random variable** if it is measurable $\mathcal{F}/\mathcal{B}(\mathbb{R})$, where $\mathcal{B}(\mathbb{R})$ is the **Borel σ-field** on $\mathbb{R}$.

Measurability condition

For every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$X^{-1}(B) = \{\omega \in \Omega : \ X(\omega) \in B\} \in \mathcal{F}.$$

Notation

We now write $\mathcal{B}(\mathbb{R})$ rather than $\mathcal{B}$, because today we also need Borel sets on $\mathbb{R}^2, \mathbb{R}^3, \ldots, \mathbb{R}^k$. By default the target σ-field on $\mathbb{R}$ is $\mathcal{B}(\mathbb{R})$, and we usually do not mention it.

Many Ways to Generate $\mathcal{B}(\mathbb{R})$

Definition and equivalent generators

$\mathcal{B}(\mathbb{R})$ is the σ-field generated by all open intervals (a, b). Earlier in the course we showed that the same σ-field is generated by each of the following classes:

- closed intervals $[a, b]$;
- half-open intervals $(a, b]$ or $[a, b)$;
- rays (a, ∞) or $(-\infty, a)$;
- rays with the endpoint included, $[a, \infty)$ or $(-\infty, a]$.

The list is not exhaustive.

Theme of today

Because $\mathcal{B}(\mathbb{R})$ can be generated in many ways, we can test measurability on a **small generating class** instead of on every Borel set. The next lemma makes this precise.

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Lemma: Pull Back a Generating Class

Lemma

Let $f : \Omega_1 \to \Omega_2$, let $\mathcal{F}_1$ be a σ-field on Ω_1, and let $\mathcal{F}_2 = \sigma(\mathcal{A})$ be the σ-field on Ω_2 generated by a class $\mathcal{A}$ of subsets of Ω_2. If

$$f^{-1}(A) \in \mathcal{F}_1 \qquad \text{for every } A \in \mathcal{A},$$

then $f^{-1}(B) \in \mathcal{F}_1$ for every $B \in \mathcal{F}_2$; that is, f is measurable $\mathcal{F}_1/\mathcal{F}_2$.

Why it matters

To prove measurability we only need to pull back the **generating sets** of the target σ-field, not every set in it.

Picture

In Ω_2, the generating class $\mathcal{A}$ sits inside $\sigma(\mathcal{A}) = \mathcal{F}_2$ (the extra sets are those needed to complete $\mathcal{A}$ to a σ-field). The lemma says: if all of $\mathcal{A}$ pulls back into $\mathcal{F}_1$, then all of $\mathcal{F}_2$ does.

Proof of the Lemma

Define the good class

Assume every $A \in \mathcal{A}$ pulls back into $\mathcal{F}_1$, and let

$$\mathcal{M} = \{B \subseteq \Omega_2 : \ f^{-1}(B) \in \mathcal{F}_1\}.$$

By assumption, $\mathcal{A} \subseteq \mathcal{M}$.

Claim: $\mathcal{M}$ is a σ-field

This is routine; check the three properties.

- $\Omega_2 \in \mathcal{M}$, since $f^{-1}(\Omega_2) = \Omega_1 \in \mathcal{F}_1$.
- If $B \in \mathcal{M}$, then $f^{-1}(B^c) = \left(f^{-1}(B)\right)^c \in \mathcal{F}_1$.
- If $B_j \in \mathcal{M}$, then $f^{-1}\left(\bigcup_j B_j\right) = \bigcup_j f^{-1}(B_j) \in \mathcal{F}_1$.

Conclude by minimality

$\mathcal{M}$ is a σ-field containing $\mathcal{A}$, and $\sigma(\mathcal{A})$ is the **smallest** such σ-field. So $\mathcal{F}_2 = \sigma(\mathcal{A}) \subseteq \mathcal{M}$, i.e. every $B \in \mathcal{F}_2$ pulls back into $\mathcal{F}_1$. □

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Theorem: Pulling Back Rays Is Enough

Theorem

Let $X : \Omega \to \mathbb{R}$. Then X is a random variable if and only if

$$\{\omega : \ X(\omega) \leq x\} = X^{-1}\big((-\infty, x]\big) \in \mathcal{F} \qquad \text{for every } x \in \mathbb{R}.$$

($\Rightarrow$)

If X is measurable, every Borel set pulls back into $\mathcal{F}$. Each $(-\infty, x]$ is a Borel set, so $\{X \leq x\} \in \mathcal{F}$ for every real x.

($\Leftarrow$)

Assume $\{X \leq x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$. The rays $(-\infty, x]$ **generate** $\mathcal{B}(\mathbb{R})$, and their pullbacks lie in $\mathcal{F}$. By the Lemma, $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$. So X is a random variable. $\square$

Corollary: Other Generating Classes

Corollary

X is a random variable if and only if $X^{-1}(I) \in \mathcal{F}$ for every I in **any** class that generates $\mathcal{B}(\mathbb{R})$, for example

$$\{(x,y)\}, \quad \{[x,y]\}, \quad \{(x,y]\}, \quad \{(x,\infty)\}, \quad \{(-\infty,x)\}, \ \ldots$$

In particular: X is a random variable $\iff \{X \in (x,y)\} \in \mathcal{F}$ for all $x < y$.

Proof

Identical to the proof of the previous theorem: the forward direction is the definition of measurability, and the reverse direction is the Lemma applied to the chosen generating class. □

Remark

The class of rays $(-\infty, x]$ is the most convenient choice, and it is the one that leads to the **distribution function** of X later in the course.

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The σ-Field Generated by a Random Variable

Definition

$\sigma(X)$ is the **smallest** σ-field on Ω with respect to which X is measurable.

Simple case (earlier lessons)

For a simple X, with finitely many values, we described $\sigma(X)$ by pulling back **arbitrary** subsets of $\mathbb{R}$:

$$\sigma(X) = \left\{X^{-1}(H) : \ H \subseteq \mathbb{R}\right\},$$

and a simple Y is $\sigma(X)$-measurable iff $Y = f(X)$ for some $f : \mathbb{R} \to \mathbb{R}$.

General case (stated today, proved next lesson)

(I) $\sigma(X) = \left\{X^{-1}(H) : \ H \in \mathcal{B}(\mathbb{R})\right\}$.

(II) Y is $\sigma(X)$-measurable iff $Y = f(X)$ for some **measurable** $f : \mathbb{R} \to \mathbb{R}$.

Changes from the simple case

H ranges over **Borel** sets only, and f must be a **measurable function**. The theorem still holds for simple variables, but it is more restrictive there.

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Vector-Valued Random Variables

Definition

A k-**dimensional random vector** is a map

$$\mathrm{X}(\omega) = \big(X_1(\omega), X_2(\omega), \ldots, X_k(\omega)\big), \qquad \mathrm{X} : \Omega \to \mathbb{R}^k.$$

The spaces involved are $(\Omega, \mathcal{F})$ and $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$.

Measurability

X is a random vector if it is measurable $\mathcal{F}/\mathcal{B}(\mathbb{R}^k)$:

$$\mathrm{X}^{-1}(B) \in \mathcal{F} \qquad \text{for every } B \in \mathcal{B}(\mathbb{R}^k).$$

What we need

To work with this definition we need a good handle on $\mathcal{B}(\mathbb{R}^k)$. First we recall Cartesian products of sets and of classes of sets.

Cartesian Products and $\mathcal{B}(\mathbb{R}^k)$

Cartesian products

For sets A, B: $A \times B = \{(a, b) : \ a \in A, \ b \in B\}$. For classes of sets $\mathcal{A}, \mathcal{B}'$:

$$\mathcal{A} \times \mathcal{B}' = \{A \times B : \ A \in \mathcal{A}, \ B \in \mathcal{B}'\}.$$

Borel σ-field on $\mathbb{R}^k$

$\mathcal{B}(\mathbb{R}^k)$ is the σ-field generated by all open rectangles

$$(a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_k, b_k).$$

If you understand $k = 2$, you understand general k.

Product σ-field

For classes $\mathcal{A}, \mathcal{B}'$, the symbol $\mathcal{A} \otimes \mathcal{B}'$ denotes the σ-field generated by $\mathcal{A} \times \mathcal{B}'$. Here

$$\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \sigma\big(\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})\big).$$

Caution

In measure theory texts, $\otimes$ on σ-fields means "generated by the Cartesian product", not a Kronecker product of matrices.

Theorem: $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$

Theorem

The Borel sets on $\mathbb{R}^2$ are exactly the σ-field generated by products of one-dimensional Borel sets:

$$\mathcal{B}(\mathbb{R}^2) = \sigma\big(\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})\big).$$

Proof of ($\subseteq$)

Take an open rectangle $(a, b) \times (c, d)$. Both (a, b) and (c, d) are in $\mathcal{B}(\mathbb{R})$, so the rectangle lies in $\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})$, hence in $\sigma\big(\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})\big)$.

$\mathcal{B}(\mathbb{R}^2)$ is the **smallest** σ-field containing all open rectangles, and $\sigma\big(\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})\big)$ is a σ-field containing them. Hence

$$\mathcal{B}(\mathbb{R}^2) \subseteq \sigma\big(\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})\big).$$

Next

The reverse inclusion takes more work. We build an auxiliary class $\mathcal{A}$ and show it is a σ-field.

Proof of ($\supseteq$), Part 1: an Auxiliary σ-Field

Define

$$\mathcal{A} = \{A \subseteq \mathbb{R} : \ A \times \mathbb{R} \in \mathcal{B}(\mathbb{R}^2)\}.$$

Claim: $\mathcal{A}$ is a σ-field on $\mathbb{R}$.

Contains $\mathbb{R}$

$\mathbb{R} \times \mathbb{R} = \mathbb{R}^2 \in \mathcal{B}(\mathbb{R}^2)$, because a σ-field on $\mathbb{R}^2$ contains the whole space. So $\mathbb{R} \in \mathcal{A}$.

Closed under complements

If $A \in \mathcal{A}$ then $A \times \mathbb{R} \in \mathcal{B}(\mathbb{R}^2)$. Since

$$A^c \times \mathbb{R} = (A \times \mathbb{R})^c,$$

and $\mathcal{B}(\mathbb{R}^2)$ is closed under complements, $A^c \times \mathbb{R} \in \mathcal{B}(\mathbb{R}^2)$, so $A^c \in \mathcal{A}$.

Closed under countable unions

If $A_n \in \mathcal{A}$, then $\left(\bigcup_n A_n\right) \times \mathbb{R} = \bigcup_n (A_n \times \mathbb{R}) \in \mathcal{B}(\mathbb{R}^2)$, so $\bigcup_n A_n \in \mathcal{A}$. Hence $\mathcal{A}$ is a σ-field. $\square$

Proof of $(\supseteq)$, Part 2: $\mathcal{A} \supseteq \mathcal{B}(\mathbb{R})$

$\mathcal{A}$ contains all open intervals

Fix $a < b$. Write the vertical strip as a countable union:

$$(a, b) \times \mathbb{R} = \bigcup_{n=1}^{\infty} (a, b) \times (-n, n).$$

Each $(a, b) \times (-n, n)$ is a generating open rectangle, so it lies in $\mathcal{B}(\mathbb{R}^2)$, and so does the countable union. Therefore $(a, b) \in \mathcal{A}$.

Conclude

$\mathcal{A}$ is a σ-field containing every open interval. $\mathcal{B}(\mathbb{R})$ is the smallest such σ-field, so $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{A}$. That is,

$$A \in \mathcal{B}(\mathbb{R}) \implies A \times \mathbb{R} \in \mathcal{B}(\mathbb{R}^2).$$

By the symmetric argument,

$$B \in \mathcal{B}(\mathbb{R}) \implies \mathbb{R} \times B \in \mathcal{B}(\mathbb{R}^2).$$

Proof of ($\supseteq$), Part 3: Finishing

Product sets are Borel

Let $A, B \in \mathcal{B}(\mathbb{R})$ be arbitrary (not necessarily intervals). Then

$$A \times B = (A \times \mathbb{R}) \cap (\mathbb{R} \times B).$$

Both sets on the right lie in $\mathcal{B}(\mathbb{R}^2)$, and a σ-field is closed under finite intersections. So $A \times B \in \mathcal{B}(\mathbb{R}^2)$.

Conclusion

We have shown $\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$. The left side generates $\sigma\big(\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})\big)$, the smallest σ-field containing it, and $\mathcal{B}(\mathbb{R}^2)$ is a σ-field containing it. Hence

$$\sigma\big(\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})\big) \subseteq \mathcal{B}(\mathbb{R}^2).$$

Together with Part 0 this gives $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. □

Intuition

A product of two wild Borel sets is the intersection of two vertical and horizontal "strips". Strips are easy to control, so products of Borel sets are Borel.

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Theorem: Components Decide

Theorem

Let $\mathrm{X} = (X_1, \ldots, X_k) : \Omega \to \mathbb{R}^k$. Then

$$\mathrm{X} \text{ is measurable } \mathcal{F}/\mathcal{B}(\mathbb{R}^k) \iff \text{each } X_i \text{ is measurable } \mathcal{F}/\mathcal{B}(\mathbb{R}).$$

In words: X is a random vector if and only if every component X_i is a random variable.

Why this is useful

The definition asks us to pull back every Borel set of $\mathbb{R}^k$. The theorem reduces this to checking one-dimensional Borel sets, one coordinate at a time.

Proof of ($\Rightarrow$)

Assume X is a random vector

Fix i and $x_i \in \mathbb{R}$. For $k = 2$ and $i = 1$:

$$\{X_1 \leq x_1\} = \mathsf{X}^{-1}\big((-\infty, x_1] \times \mathbb{R}\big).$$

In general,

$$\{X_i \leq x_i\} = \mathsf{X}^{-1}\big(\mathbb{R} \times \cdots \times \mathbb{R} \times (-\infty, x_i] \times \mathbb{R} \times \cdots \times \mathbb{R}\big).$$

Conclude

The set inside $\mathsf{X}^{-1}(\cdot)$ is a Borel set of $\mathbb{R}^k$, and X is measurable, so $\{X_i \leq x_i\} \in \mathcal{F}$ for every real x_i. By the earlier theorem (pull back rays), X_i is a random variable. □

Picture ($k = 2$)

The points ω with $X_1(\omega) \leq x_1$ are exactly the points that X sends into the half-plane $(-\infty, x_1] \times \mathbb{R}$.

Proof of ($\Leftarrow$)

Assume every X_i is measurable

$\mathcal{B}(\mathbb{R}^k)$ is generated by the sets $(-\infty, x_1] \times \cdots \times (-\infty, x_k]$, so by the Lemma it is enough to show their pullbacks lie in $\mathcal{F}$.

Pullback of a generating set

A point ω is sent into this set exactly when **all** coordinates land in the right place at the same time:

$$X^{-1}\big((-\infty, x_1] \times \cdots \times (-\infty, x_k]\big) = \bigcap_{i=1}^{k} \{X_i \leq x_i\} = \bigcap_{i=1}^{k} {X_i}^{-1}\big((-\infty, x_i]\big).$$

Conclude

Each ${X_i}^{-1}\big((-\infty, x_i]\big) \in \mathcal{F}$ because X_i is measurable. A σ-field is closed under finite intersections, so the right side lies in $\mathcal{F}$. By the Lemma, $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{B}(\mathbb{R}^k)$. $\square$

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Summary

Key results

Concept / Result	Content
Measurable map	$f^{-1}(B) \in \mathcal{F}_1$ for all $B \in \mathcal{F}_2$ (measurable $\mathcal{F}_1/\mathcal{F}_2$)
Random variable	measurable $\mathcal{F}/\mathcal{B}(\mathbb{R})$
Lemma	if $\mathcal{F}_2 = \sigma(\mathcal{A})$ and $f^{-1}(\mathcal{A}) \subseteq \mathcal{F}_1$, then f is measurable $\mathcal{F}_1/\mathcal{F}_2$
Rays criterion	X is a random variable $\iff \{X \leq x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$
$\sigma(X)$ (general)	$\{X^{-1}(H) : H \in \mathcal{B}(\mathbb{R})\}$; $Y = f(X)$ with f measurable
Borel sets on $\mathbb{R}^2$	$\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$
Random vector	X measurable $\iff$ each component X_i measurable

Exercises

1. Prove that $\mathcal{M} = \{B \subseteq \Omega_2 : \ f^{-1}(B) \in \mathcal{F}_1\}$ is a σ-field (the claim in the Lemma's proof).
2. Show that X is a random variable if and only if $\{X < x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$.
3. Show that X is a random variable if and only if $\{X \in (x, y)\} \in \mathcal{F}$ for all $x < y$.
4. Verify that $(A \times B)^c$ need not be a product set, and write it as a union of two product sets.
5. Prove the symmetric statement: if $B \in \mathcal{B}(\mathbb{R})$ then $\mathbb{R} \times B \in \mathcal{B}(\mathbb{R}^2)$.
6. Extend the proof of $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$ to $k = 3$.
7. Let $\mathrm{X} = (X_1, X_2)$ with X_1, X_2 random variables. Show that $\{X_1 \leq x_1, \ X_2 \leq x_2\} \in \mathcal{F}$ for all $x_1, x_2 \in \mathbb{R}$.

Outlook

Next lesson

We prove the theorem stated today without proof: the description of $\sigma(X)$ through Borel sets, and the relationship between the σ-field generated by (X_1, X_2) and the σ-fields $\sigma(X_1)$, $\sigma(X_2)$ generated individually.

Looking ahead

After that we move to **sequences of random variables** and **almost sure convergence**.